

KUCHUN VENKATA SANJEEV

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Computer Science undergraduate (Honors through Research Track) with full-stack development experience in Java and Spring Boot, RESTful API design, and SQL-backed systems. Applied research background in deep learning and Explainable AI. Strong foundation in data structures, algorithms, and object-oriented design; authored multiple research papers with two papers are currently under peer review and additional manuscripts are submitted for publication.

EDUCATION

Koneru Lakshmaiah Education Foundation (KL University) — Guntur, India Expected May 2028

B. Tech, Computer Science and Engineering — Honors through Research Track | CGPA: 8.9/10

TECHNICAL SKILLS

Languages: Java, C, SQL

Backend & APIs: Spring Boot, Microservices, Hibernate, RESTful API design, JWT Authentication

Frontend: React JS, JavaScript, HTML/CSS

Databases: MySQL, PostgreSQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Full-Stack Development

Tools & Cloud: Git, Maven, AWS, Postman, Swagger/OpenAPI

RESEARCH EXPERIENCE

AI-Guided Thoracic Abnormality Detection Using Grad-CAM — KL University 2025

- Built CNN-based deep learning models in Python for thoracic abnormality detection from chest X-ray images, reaching 93.5% accuracy in pneumonia detection.
- Developed novel Grad-CAM visualization techniques that directly supported clinical trust in AI-driven diagnostic tools, reducing model interpretation time by 30%. Authored and submitted a research paper on AI-guided thoracic abnormality detection to Springer's Journal of Signal, Image and Video Processing.

PROJECTS

Online Service Management System | React, Spring Boot, MySQL 2026

- Built a full-stack platform with JWT authentication, RESTful APIs, and role-based access control.
- Designed and deployed responsive React interfaces, integrating seamlessly with MySQL databases through implemented CRUD operations, enabling real-time data synchronization for 500+ users.

Automatic Fire Alarm System for Indian Railway Coaches | Arduino, Embedded Systems 2024–2025

- Designed an automated fire detection and alert system using flame sensors and servo motors, achieving ~98% detection accuracy.

PUBLICATIONS

- Kuchun, V.S., "Unrevealing the Black Box Property of AI-Guided Thoracic Abnormality Detection Using Grad-CAM." Submitted to Springer (With Editor); **Indian Patent Published**, 2026, **Application No. [202641107116]**.
- Kuchun, V.S., "AI-Driven 3D Digital Twin Framework for Energy Damage Detection and Predictive Recovery in Green Hydrogen Pipelines." (Under peer review)

LEADERSHIP & CERTIFICATIONS

- Director, KL-VEDA (Research & Discovery Wing), KL University (2025–Present) — lead student-driven AI research initiatives and mentor undergraduate researchers.
- Joint Secretary, Documentation & Publications Division – Student Activity Centre (SAC), KL University (2026–Present)
- Certifications: AWS Cloud Practitioner Foundations · GitHub Copilot Certification · Certified C Developer (W3Schools) · CEFR B1 English.